

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B. Tech. (CE/IT/EE/CL/EC/ME) Examination

December 2010

ME102 Fundamental of Mechanical Engineering

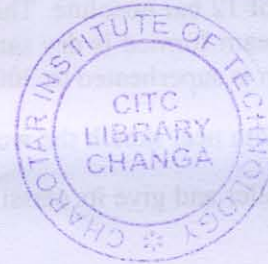
Date: 10.12.2010 , Friday

Time: 10:30 a.m to 1:30 p.m

Maximum Marks: 70

Instructions:

1. Attempt all the questions.
2. Figures to the right indicate full marks.
3. Make suitable assumptions wherever necessary.
4. Draw neat sketch wherever required.



FOR REFERENCE

SECTION- I

- Q1(a) What is Fuel and combustion? Explain HCV and LCV of fuel. 3
- (b) Write advantages of non-conventional sources of energy over conventional. 4
- (c) Write similarities and dissimilarities of heat and work; Also give its sign conventions. 3
- Q2(a) State various Non flow processes. Explain constant pressure process in detail. 5
- (b) A tank contains 3m^3 of air at 25 bar absolute pressure. This air is cooled until its pressure and temperature decrease to 15 bar and 21°C respectively. Determine change in Internal Energy, Change in Enthalpy, Heat Transferred and Work done. Take $C_p=1.005\text{ KJ/kg K}$ and $C_v=0.718\text{ KJ/kg K}$ for air. 5

OR

- Q2(a) Define Vapour, Gas and Perfect Gas. Explain with graph Boyle and Gay-Lussac law. 5
- (b) Initial volume of 0.18 kg of certain gas was 0.15m^3 at a temperature 15°C and pressure 1 bar. After adiabatic compression to 0.056m^3 , the pressure was found 4 bar. Find (i) Gas constant, (ii) Molecular mass, (iii) γ , (iv) C_p and C_v (v) Change in internal energy. 5
- Q3(a) Derive the Air standard efficiency of Otto cycle. 5
- (b) Explain with neat sketch working of 4 stroke diesel engine. What is Thermal efficiency? 5
- (c) The engine working on ideal Otto cycle. The temperature at the beginning and at the end of the compression is 50°C and 400°C . Calculate the compression ratio and air standard efficiency. 5

OR

- Q3(a) Differentiate Gasoline Engine and Diesel Engine. 5
- (b) Explain working of 2 stroke petrol engine. What is Mechanical Efficiency? 5
- (c) The compression ratio of an oil engine working on Diesel cycle is 15. Cut off takes place at 12% of the working stroke. The air is drawn in to cylinder at 100 kPa and 27°C . Assume $C_p=1.006\text{ KJ/kg K}$ and $C_v=0.717\text{ KJ/kg K}$ Calculate: (i) Temperature at the end of the compression, (ii) Pressure at the end of compression, (iii) Air standard efficiency of the cycle. 5

SECTION II

- Q4(a) Prove that Dryness fraction + Wetness fraction = 1 4
- (b) Explain different states of steam. 3
- (c) State various mountings and accessories used in Boilers. 3
- Q5(a) Calculate the heat supplied to the boiler/kg of steam when the steam is generated at constant pressure of 12 bar absolute. The water is supplied to the boiler is 50°C and at the same pressure. If (i) The steam formed is dry saturated, (ii) The steam is wet with the dryness fraction of 0.91. (iii) The steam is superheated to 300°C. Take $C_p = 2.1 \text{ kJ/kg K}$. 5
- (b) Explain with neat sketch the working of Cochran boiler. 5
- (c) Define boiler and give its classification and specification. 5

OR

- Q5(a) Explain Construction and working of Babcock and Wilcox boiler with neat sketch. 6
- (b) A boiler produces 1000 kg of wet steam per hour at a pressure of 8 bar. The dryness of steam is 0.95, Feed water temperature is 10°C. If boiler efficiency is 75%, calculate the amount of coal consumed per hour. Calorific value for coal is 32,500 kJ/kg. 4
- (c) Explain Vapour compression refrigeration with neat sketch and Differentiate refrigeration and air conditioning. 5
- Q6(a) Classify Air compressors. Explain working of reciprocating Air compressor with neat sketch. 5
- (b) A single cylinder, single acting air compressor has a cylinder diameter 160mm and stroke length 300mm. It draws air into its cylinder at a pressure of 1 bar and temperature 27°C. This air is then compressed to a pressure of 6.5 bar. If the compressor runs at a speed of 120 rpm Calculate, (1) Mass of air compressed/cycle, (2) Work required/cycle, (3) power required to drive the compressor if the compression is adiabatic. 5

OR

- Q6(a) Classify pumps and Explain vortex type centrifugal pump with neat sketch. 5
- (b) State various power transmission drives And explain its relative merits and demerits. 5

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